In [3]:
```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from sklearn.datasets import fetch_california_housing
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.model_selection import train_test_split

# 1. Load the California Housing Dataset
housing = fetch_california_housing(as_frame=True)
df = housing.frame

# Note: If using your own local CSV file instead, replace the above lines with:
# df = pd.read_csv('house_prices.csv')

# 2. Data Preprocessing & Feature Selection
# Handle missing values if any
df = df.dropna()

# Define independent variables (features) and dependent variable (target)
X = df.drop(columns=["MedHouseVal"])
y = df["MedHouseVal"]

# 3. Split Dataset into Training and Testing Sets (80:20 ratio)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 4. Train the Linear Regression Model
model = LinearRegression()
model.fit(X_train, y_train)

# 5. Make Predictions on the Test Set
y_pred = model.predict(X_test)

# 6. Evaluate Model Performance Metrics
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("--- Model Performance Evaluation ---")
print(f"Mean Absolute Error (MAE): {mae:.4f}")
print(f"Mean Squared Error (MSE): {mse:.4f}")
print(f"Root Mean Squared Error (RMSE): {rmse:.4f}")
print(f"R² Score: {r2:.4f}")

# 7. Analyze Regression Coefficients
coefficients = pd.DataFrame(
    model.coef_, X.columns, columns=["Coefficient Value"]
)
print("\n--- Regression Coefficients ---")
print(coefficients)

# 8. Visualize Actual vs. Predicted Values
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred, alpha=0.3, color="teal")
plt.plot(
```

```
    [y_test.min(), y_test.max()],
    [y_test.min(), y_test.max()],
    color="crimson",
    linestyle="--",
    lw=2,
    label="Ideal Fit",
)
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Actual vs. Predicted House Prices (Linear Regression)")
plt.legend()
plt.show()
```

```
--- Model Performance Evaluation ---
Mean Absolute Error (MAE): 0.5332
Mean Squared Error (MSE): 0.5559
Root Mean Squared Error (RMSE): 0.7456
R² Score: 0.5758

--- Regression Coefficients ---
            Coefficient Value
MedInc               0.448675
HouseAge             0.009724
AveRooms            -0.123323
AveBedrms            0.783145
Population          -0.000002
AveOccup            -0.003526
Latitude            -0.419792
Longitude           -0.433708
```

